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## Key research and advancements

The promising characteristics of mesoporous silicon have prompted extensive research in academia and industry.

Researchers are investigating how it can improve LED efficiency, enhance photodetectors, and enable quantum-dot-based displays. Its ability to control light propagation at the nanoscale is particularly relevant for next-generation optical computing.

Due to its biocompatibility, mesoporous silicon is being studied for drug delivery systems, tissue engineering, and biosensors. The porous structure can hold therapeutic agents and release them controlled, improving the efficacy of targeted drug delivery.

In energy storage, mesoporous silicon shows promise in

next-generation batteries and supercapacitors. Its large surface area enhances ion transport and energy density. Some studies suggest silicon-based anodes in lithium-ion batteries could significantly extend their lifespan and capacity.

One of the industry's ongoing challenges is reducing material waste and energy use. Mesoporous silicon, fabricated at lower temperatures and with higher efficiency, offers a more sustainable alternative to conventional silicon processing.

## Impact on the semiconductor industry

Incorporating mesoporous silicon into semiconductor technologies can result in faster, more energy-efficient, and environmentally friendly electronics. Its inclusion in transistors can improve electron mobility and reduce power consumption.

The lightweight and porous

characteristics of m-Si make it suitable for flexible, stretchable electronics used in wearable devices. Solar panels employing mesoporous silicon could harvest sunlight more effectively and generate electricity with higher efficiency.

Its tunable porosity is ideal for ultra-sensitive gas, chemical, and biomedical sensors.

Mesoporous silicon represents a significant shift in semiconductor material development, offering unmatched control over electronic, optical, and chemical behaviour. As research progresses and industrial applications materialise, this material may redefine microelectronics, energy storage, and biomedical engineering. Despite ongoing challenges, its potential to deliver faster, more sustainable, and higher-performance semiconductor devices places mesoporous silicon at the forefront of modern technology. **EFY**